

METROLOGYAI

SIH 2026 PPT Content Guide

Problem Statement ID: PS26034

Team ID: 57893 | Team Name: Zillion MindsSashakt

Theme: Software | Category: Software

This document contains all content for the SIH 2026 presentation. Each section maps to a specific slide. Follow the placement instructions for flowcharts, diagrams, and visual elements.

SLIDE MAP: Slide 1 (Title) → Slide 2 (Problem/Solution/Unique) → Slide 3 (Technical) → Slide 4 (Feasibility) → Slide 5 (Impact) → Slide 6 (References)

SLIDE 1: TITLE PAGE

CONTENT TO PASTE:

Field	Content
Problem Statement ID	PS26034
Problem Statement Title	AI-Assisted Legal Metrology Compliance Inspection System
Theme	Software
PS Category	Software
Team ID	57893
Team Name	Zillion MindsSashakt

PLACEMENT INSTRUCTIONS:

- Use the existing SIH template layout (hexagons, SIH brain logo already present)
- Replace 'Problem Statement ID –' with 'Problem Statement ID – PS26034'
- Replace 'Team ID-' with 'Team ID – 57893'
- Replace 'Team Name' with 'Team Name – Zillion MindsSashakt'

SLIDE 2: PROBLEM + SOLUTION + UNIQUE SOLUTIONS

TITLE: METROLOGYAI

SECTION A: Understanding the Problem (Left Column)

— Paste as bullet points —

- 01 — Manual Inspection: Inspectors manually examine packaged commodity labels, making repetitive checks time-consuming.
- 02 — Multi-Surface Declarations: Required information may appear on different sides of the package.
- 03 — Real-World Image Issues: Blur, glare, perspective distortion, and small text reduce readability.
- 04 — Uncertain AI Results: A system may confuse an unreadable declaration with a genuinely missing one.
- 05 — Fragmented Verification: Physical packaging, online listings, and historical data need separate comparison.

QUOTE (at bottom of problem section):

"The challenge is not simply reading a label — it is producing a reliable, explainable inspection decision."

SECTION B: Our Solution (Center Column)

— Paste as bullet points —

- Takes product images (multi-view) → AI reads all visible text → Extracts 11 mandatory declaration fields
- Validates each field against 11 Legal Metrology rules
- Calculates compliance score (0-100) with evidence chain for every finding
- Detects label anomalies (tampering, overlays, altered MRP)
- Prioritizes inspection risk (low / medium / high)
- Inspector reviews, corrects, and makes final decision
- Saves all data to database and generates digital PDF report

QUOTE (at bottom of solution section):

"AI assists the inspection; the inspector makes the final decision."

SECTION C: Our Unique Solutions (Right Column)

— Paste as compact cards —

Multi-View Intelligence: Analyzes front, back, side, top, and bottom package views simultaneously.

Missing ≠ Unreadable: Separates absent declarations from low-confidence detection.

Adaptive Re-Capture: Identifies poor image quality and recommends specific improvements.

Evidence Chain: Links every finding to image evidence, OCR text, field, and rule.

Risk Prioritization: AI scores every inspection for risk level (low/medium/high).

Human-in-the-Loop: Inspector corrections trigger automatic smart revalidation.

DIAGRAM: USP Cards (Right Side)

Create 4 rounded rectangle cards arranged in a 2×2 grid on the right side of the slide:

- Card 1 (Top-Left): Real-time compliance scoring

- Card 2 (Top-Right): Evidence-backed findings for every violation
- Card 3 (Bottom-Left): Multi-view package analysis
- Card 4 (Bottom-Right): Inspector corrections with auto-revalidation

Use light blue fill with colored borders. Label above: "USPs {"

SLIDE 3: TECHNICAL APPROACH

TITLE: TECHNICAL APPROACH

SECTION A: Methodology (Top — Horizontal Process)

Create a horizontal flow with 7 numbered steps connected by arrows:

Step	Action	Description
01	CAPTURE	Multi-view product images
02	QUALITY CHECK	Blur + glare + visibility analysis
03	EXTRACT	AI Vision + OCR via Gemini API
04	STRUCTURE	11 declaration fields extracted
05	VALIDATE	11 Legal Metrology rules checked
06	EVIDENCE	Findings linked to source + rule
07	REPORT	Inspector review + PDF report

SECTION B: Technical Approach (Left — Bullets)

- Image Acquisition — Capture product package images and optional e-commerce listing screenshots.
- Image Quality Analysis — Client-side canvas: blur (Laplacian variance), brightness, glare (bright pixel ratio), resolution, aspect ratio.
- AI Vision + OCR — Google Gemini Vision API extracts 11 fields with confidence scores (0-100). Multi-model fallback: tries 4 models if one fails.
- Structured Extraction — 40+ field name aliases normalized. Fields: Product Name, Manufacturer, Net Quantity, MRP, Consumer Care, Dates, Country, Batch, FSSAI, Veg/Non-Veg.
- Compliance Engine — 11 rules with severity levels. Weighted scoring across 7 categories. Status: Compliant (≥80) / Review Required / Non-Compliant.
- Evidence Engine — Every finding linked to: Image Region → OCR Text → Field → Rule → Validation → Decision.

SECTION C: Tech Stack (Bottom)

Layer	Technology
Frontend	React 19 + TypeScript + Vite + Tailwind CSS + shadcn/ui + Lucide Icons
Backend	Convex (Serverless) — Actions, Mutations, Queries
Database	Convex DB — 3 tables: users, inspections, ecommerceComparisons
AI / Vision	Google Gemini Vision API (4-model fallback, temp 0.1, 4096 tokens)
Auth	Convex Auth — Email OTP + Anonymous Login
Analytics	Recharts (bar, pie, line charts)
Deployment	Freebuff Platform (auto-deploy)

DIAGRAM: Pipeline Flowchart (Right Side)

Create a vertical flowchart on the right side with 7 rounded rectangles connected by down arrows:

IMAGE → QUALITY CHECK → AI VISION + OCR → RULE VALIDATION → SCORE + EVIDENCE →
INSPECTOR REVIEW → PDF REPORT

Use different colors: Light Blue (start), Light Green (quality), Light Orange (AI), Light Red (rules), Light Blue (score), Light Green (review), Light Orange (report). Connect with navy down arrows.

SLIDE 4: FEASIBILITY AND VIABILITY

SECTION A: Feasibility Analysis (Left Column)

#	Area	Analysis
01	Technology	Gemini Vision API is production-ready. React + Convex are proven frameworks. No experimental tech.
02	Data	Product label images are easy to collect. 11 mandatory declarations publicly documented in Legal M...
03	Architecture	AI extraction and rule validation are modular, independently testable components. Clean API bound...
04	Deployment	Web-based — works on desktop, tablet, mobile. No app install. Cloud-hosted. Auto-deployment on...
05	Scalability	New categories added by updating RULE_REQUIREMENTS config. New fields added through cont...

SECTION B: What Ifs...? — Challenge → Mitigation (Center)

Challenge	Mitigation
AI misreads text	Confidence scoring (0-100) + evidence chain + inspector correction
Image is blurry	Client-side quality detection + adaptive recapture recommendations
Declaration on another side	Multi-view analysis (up to 6 package views)
Regulations change	Version-controlled rule engine — update rules without code changes
AI produces false finding	Human-in-the-loop verification — inspector makes final call
API fails	Multi-model fallback (4 Gemini models tried in sequence)

SECTION C: Before MetrologyAI → After MetrologyAI (Bottom)

	Before MetrologyAI	After MetrologyAI
Reading	Manual label reading	AI-assisted OCR + Vision extraction
Validation	Manual rule checking	Automated 11-rule compliance engine
Evidence	Scattered, paper-based	Evidence-linked digital findings
Reporting	Manual documentation	Structured digital PDF reports
Risk	Inspector-led prioritization	AI-scored risk assessment

SLIDE 5: IMPACT AND BENEFITS

SECTION A: Inspector Benefits (Left Column)

- Faster inspection — AI reads 11 fields in seconds, not minutes
- Better accuracy — Confidence scores show what is reliable
- Complete evidence — Every finding linked to source image and rule
- Risk prioritization — Focus on high-risk products first
- Smart revalidation — Corrections automatically update compliance status
- Audit trail — Complete record of every decision made

SECTION B: Operational Benefits (Center Column)

- Consistent checking — Same 11 rules applied every time
- Reduced false positives — Missing vs unreadable distinction
- Scalable — Supports more inspections without proportional staff increase
- Data-driven — Analytics show patterns across inspections
- Versioned rules — New categories added through configuration
- Multi-view analysis — One inspection covers entire package

SECTION C: Social & Economic Benefits (Right Column)

- Better consumer protection through consistent enforcement
- Faster detection of non-compliant products
- Digital evidence supports legal proceedings
- Prevents consumers from buying mislabeled products
- Supports government Digital India initiative
- Transparent, explainable AI decisions

DIAGRAM: Impact Bar Chart (Bottom or Right)

Create a horizontal bar chart with 5 impact dimensions:

Dimension	Proportional %	Color
Inspection Efficiency	35%	Green
Evidence & Traceability	25%	Blue
Inspector Assistance	20%	Saffron/Orange
Consistency	10%	Navy
Digital Documentation	10%	Gray

Note: Label as "Illustrative impact priorities" — do NOT claim exact percentages as measured results.

SLIDE 6: RESEARCH AND REFERENCES

SECTION A: Legal & Regulatory References (Left Column)

- 01 — Legal Metrology Act, 2009 — Government of India legislation governing weights and measures
- 02 — Legal Metrology (Packaged Commodities) Rules, 2011 — Mandatory declarations on pre-packaged goods
- 03 — Department of Consumer Affairs — Legal Metrology Division (legalmetrology.nic.in)
- 04 — Official Gazette Notifications — Amendments and updates to packaged commodity rules

SECTION B: Research Areas (Center Column)

- Computer Vision & OCR — Scene-text recognition for product label analysis
- Multimodal AI — Vision-language models for regulatory compliance
- Explainable AI — Confidence scoring and evidence-based decision support
- Human-AI Interaction — Inspector-in-the-loop verification patterns

SECTION C: Technology References (Right Column)

- React — react.dev
- TypeScript — typescriptlang.org
- Vite — vitejs.dev
- Tailwind CSS — tailwindcss.com
- Convex — convex.dev
- Google Gemini Vision API — ai.google.dev
- Recharts — recharts.org
- shadcn/ui — ui.shadcn.com

PLACEHOLDERS (Bottom)

- [ADD VERIFIED RESEARCH PAPERS OR ACADEMIC REFERENCES HERE]
- [ADD ANY ADDITIONAL TECHNOLOGY DOCUMENTATION LINKS HERE]

Note: Do NOT fabricate paper titles, authors, statistics, or URLs.

IMPORTANT NOTES FOR PPT CREATION

Preserve SIH Template

Keep the blue footer, team name oval, SIH logo, hexagons, and slide dimensions unchanged.

Editable Shapes

All flowcharts, diagrams, and tables must be created as editable PowerPoint shapes, not images.

No AI Remarks

Do NOT include any "Generated by AI" or tool branding.

No Fake Claims

Do NOT invent accuracy percentages, processing times, or cost savings.

Legal Disclaimer

Describe as "AI-assisted inspection and decision-support system" — NOT as official government authority.

Readability

Font sizes must be readable when projected. Avoid tiny text. Short sentences > paragraphs.

Alignment

Check every slide for equal spacing, consistent margins, no overlapping elements.

File Format

Save as .pptx (editable) and upload as PDF to the SIH portal.